Universidad San Jorge

Task #4 - Roman Converter - Test Automation

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1. **Successful Tests**

The tests that have been verified to output the expected, correct results are:

- Limit values (1, 2, 3998, 3999)
- 6 subtractive pairs: IV, IX, XL, XC, CD, CM
- Complex numbers (1994→MCMXCIV, 3888→MMMDCCCLXXXVIII)
- Exact symbol values (5→V, 10→X, 50→L, 100→C, 500→D, 1000→M)
- Invalid input validation:
    - Out of bounds numbers(0, -1, 4000)
    - Invalid Roman numbers ("IIII", "VV", "IC", "abc", "")
- Lower caps to Upper caps conversión ("mcmxciv"→1994)
- Consistent round tripping (int→roman→int = initial value)

2. **Failed tests**

The failed tests that reveal an important limitation in the integerToRoman() function

- TC-I-24: integerToRoman(3.5) Decimals not rejected
- TC-AI-I-06: integerToRoman(2.7) Decimals not rejected
- TC-AI-I-08: integerToRoman(undefined) Undefined values are not rejected
- TC-AI-I-09: integerToRoman("5") Improper strings are not rejected or converted to integers

The function only validates the range from 1 to 3999, but the type of the input variable is not validated. If a decimal, string or undefined value are input, the function tries to process it as opposed to rejecting it

These cases are coding errors where the function should validate the input before processing it.

3. **Screenshots:**

SET 1: Manual Test Cases - integerToRoman

Boundary Value Analysis (BVA)

- ✓ TC-I-01: should convert 1 to "I" (minimum valid value)
- ✓ TC-I-02: should convert 2 to "II" (near minimum)
- ✓ TC-I-19: should convert 3998 to "MMMCMXCVIII" (near maximum)
- ✓ TC-I-20: should convert 3999 to "MMMCMXCIX" (maximum valid value)

Equivalence Partitioning (EP)

- ✓ TC-I-03: should convert 3 to "III" (triple repetition of I)
- ✓ TC-I-05: should convert 5 to "V" (exact symbol value)
- ✓ TC-I-07: should convert 10 to "X" (exact symbol value)
- ✓ TC-I-10: should convert 50 to "L" (exact symbol value)
- ✓ TC-I-12: should convert 100 to "C" (exact symbol value)
- ✓ TC-I-14: should convert 500 to "D" (exact symbol value)
- ✓ TC-I-16: should convert 1000 to "M" (exact symbol value)
- ✓ TC-I-17: should convert 1994 to "MCMXCIV" (complex mid-range)
- ✓ TC-I-18: should convert 3888 to "MMMDCCCLXXXVIII" (longest valid numeral)

Subtractive Pairs Coverage

- ✓ TC-I-04: should convert 4 to "IV" (subtractive pair IV)
- ✓ TC-I-06: should convert 9 to "IX" (subtractive pair IX)
- ✓ TC-I-09: should convert 40 to "XL" (subtractive pair XL)
- ✓ TC-I-11: should convert 90 to "XC" (subtractive pair XC)
- ✓ TC-I-13: should convert 400 to "CD" (subtractive pair CD)
- ✓ TC-I-15: should convert 900 to "CM" (subtractive pair CM)
- ✓ TC-I-08: should convert 14 to "XIV" (combined X + IV)

```
expect(integerToRoman(14)).to.equal('XIV');
```

Invalid Inputs

- ✓ TC-I-21: should throw error for 0 (below minimum)
- ✓ TC-I-22: should throw error for -1 (negative number)
- ✓ TC-I-23: should throw error for 4000 (above maximum)
- ✗ TC-I-24: should throw error for 3.5 (non-integer)

```
AssertionError: expected [Function] to throw an error
    at n.<anonymous> (tests.js:108:49)
```

SET 1: Manual Test Cases - romanToInteger

Boundary Value Analysis (BVA)

- ✓ TC-R-01: should convert "I" to 1 (minimum valid numeral)
- ✓ TC-R-02: should convert "II" to 2 (near minimum)
- ✓ TC-R-18: should convert "MMMCMXCVIII" to 3998 (near maximum)
- ✓ TC-R-19: should convert "MMMCMXCIX" to 3999 (maximum valid numeral)

Equivalence Partitioning (EP)

- ✓ TC-R-03: should convert "III" to 3 (triple repetition)
- ✓ TC-R-05: should convert "V" to 5 (single symbol)
- ✓ TC-R-07: should convert "X" to 10 (single symbol)
- ✓ TC-R-09: should convert "L" to 50 (single symbol)
- ✓ TC-R-11: should convert "C" to 100 (single symbol)
- ✓ TC-R-13: should convert "D" to 500 (single symbol)
- ✓ TC-R-15: should convert "M" to 1000 (single symbol)
- ✓ TC-R-16: should convert "MCMXCIV" to 1994 (complex mid-range)
- ✓ TC-R-17: should convert "MMMDCCCLXXXVIII" to 3888 (longest numeral)

Subtractive Pairs Coverage

- ✓ TC-R-04: should convert "IV" to 4 (subtractive pair IV)
- ✓ TC-R-06: should convert "IX" to 9 (subtractive pair IX)
- ✓ TC-R-08: should convert "XL" to 40 (subtractive pair XL)
- ✓ TC-R-10: should convert "XC" to 90 (subtractive pair XC)
- ✓ TC-R-12: should convert "CD" to 400 (subtractive pair CD)
- ✓ TC-R-14: should convert "CM" to 900 (subtractive pair CM)

Invalid Inputs

- ✓ TC-R-20: should throw error for "IIII" (invalid repetition)
- ✓ TC-R-21: should throw error for "VV" (V cannot repeat)
- ✓ TC-R-22: should throw error for "IC" (invalid subtractive)
- ✓ TC-R-23: should throw error for empty string
- ✓ TC-R-24: should throw error for "abc" (non-Roman characters)
- ✓ TC-R-25: should convert "mcmxciv" to 1994 (lowercase auto-conversion)

SET 1: Round-Trip Consistency Tests

- ✓ TC-RT-01: should maintain consistency for 1 → "I" → 1
- ✓ TC-RT-02: should maintain consistency for 1994 → "MCMXCIV" → 1994
- ✓ TC-RT-03: should maintain consistency for 3999 → "MMMCMXCIX" → 3999
- ✓ TC-RT-04: should maintain consistency for "XIV" → 14 → "XIV"
- ✓ TC-RT-05: should maintain consistency for "MMMCMXCIX" → 3999 → "MMMCMXCIX"

SET 2: AI-Generated Test Cases - integerToRoman

Boundary Tests

- ✓ TC-AI-I-01: should convert 1 to "I"
- ✓ TC-AI-I-02: should convert 3999 to "MMMCMXCIX"

Invalid Inputs - Comprehensive

- ✓ TC-AI-I-03: should throw error for 0
- ✓ TC-AI-I-04: should throw error for -5
- ✓ TC-AI-I-05: should throw error for 4000
- ✗ TC-AI-I-06: should throw error for 2.7 (non-integer)

```
AssertionError: expected [Function] to throw an error
    at n.<anonymous> (tests.js:289:49)
```

- ✓ TC-AI-I-07: should throw error for null
- ✗ TC-AI-I-08: should throw error for undefined

```
AssertionError: expected [Function] to throw an error
    at n.<anonymous> (tests.js:297:55)
```

- ✗ TC-AI-I-09: should throw error for "5" (string instead of number)

```
AssertionError: expected [Function] to throw an error
    at n.<anonymous> (tests.js:301:49)
```

Subtractive Pairs - All Six

Subtractive Pairs - All Six

- ✓ TC-AI-I-10: should convert 4 to "IV"
- ✓ TC-AI-I-11: should convert 9 to "IX"
- ✓ TC-AI-I-12: should convert 40 to "XL"
- ✓ TC-AI-I-13: should convert 90 to "XC"
- ✓ TC-AI-I-14: should convert 400 to "CD"
- ✓ TC-AI-I-15: should convert 900 to "CM"
- ✓ TC-AI-I-16: should convert 44 to "XLIV" (XL + IV combined)
- ✓ TC-AI-I-17: should convert 99 to "XCIX" (XC + IX combined)
- ✓ TC-AI-I-18: should convert 1999 to "MCMXCIX" (multiple subtractives)

Historical and Contextual Cases

- ✓ TC-AI-I-19: should convert 2024 to "MMXXIV" (recent year)
- ✓ TC-AI-I-20: should convert 3888 to "MMMDCCCLXXXVIII" (longest)
- ✓ TC-AI-I-21: should convert 500 to "D" (single symbol)
- ✓ TC-AI-I-22: should convert 1500 to "MD" (two symbols)
- ✓ TC-AI-I-23: should convert 3000 to "MMM" (max M repetition)
- ✓ TC-AI-I-24: should convert 58 to "LVIII" (mid-range additive)
- ✓ TC-AI-I-25: should convert 1776 to "MDCCLXXVI" (historical year)

SET 2: AI-Generated Test Cases - romanToInteger

Boundary Tests

- ✓ TC-AI-R-01: should convert "I" to 1
- ✓ TC-AI-R-02: should convert "MMMCMXCIX" to 3999

Invalid Inputs - Extended Coverage

- ✓ TC-AI-R-03: should throw error for empty string
- ✓ TC-AI-R-04: should throw error for "IIII" (4 consecutive I)
- ✓ TC-AI-R-05: should throw error for "VV" (V cannot repeat)
- ✓ TC-AI-R-06: should throw error for "LL" (L cannot repeat)
- ✓ TC-AI-R-07: should throw error for "DD" (D cannot repeat)
- ✓ TC-AI-R-08: should throw error for "IC" (invalid subtractive)
- ✓ TC-AI-R-09: should throw error for "IL" (invalid subtractive)
- ✓ TC-AI-R-10: should throw error for "VX" (invalid subtractive)
- ✓ TC-AI-R-11: should throw error for "abc" (non-Roman chars)

Complex Combinations

- ✓ TC-AI-R-12: should convert "MCMXCIV" to 1994
- ✓ TC-AI-R-13: should convert "MMXXIV" to 2024
- ✓ TC-AI-R-14: should convert "MDCCLXXVI" to 1776

Subtractive Pairs

- ✓ TC-AI-R-15: should convert "IV" to 4
- ✓ TC-AI-R-16: should convert "IX" to 9
- ✓ TC-AI-R-17: should convert "XL" to 40
- ✓ TC-AI-R-18: should convert "XC" to 90
- ✓ TC-AI-R-19: should convert "CD" to 400
- ✓ TC-AI-R-20: should convert "CM" to 900
- ✓ TC-AI-R-21: should convert "XLIV" to 44 (combined XL + IV)
- ✓ TC-AI-R-22: should convert "XCIX" to 99 (combined XC + IX)

Edge Cases

- ✓ TC-AI-R-23: should convert "mcmxciv" to 1994 (lowercase)
- ✓ TC-AI-R-24: should convert "MMMDCCCLXXXVIII" to 3888 (longest)
- ✓ TC-AI-R-25: should convert "MMMCMXCVIII" to 3998 (near maximum)

Test Suite Summary

- ✓ Total test count: should have 100+ tests across both sets